

python : 3.13.12 on Windows AMD64
torch : 2.13.0+cpu numpy: 2.4.3 matplotlib: 3.10.8
backend : CPU (cpu)
seed : 0
Training set has 1400 samples
Validation set has 300 samples
Test set has 300 samples
alpha is 0.1
omega is 1
model: 4,353 parameters, 3 Linear layers

training: Adam(lr=0.002), cosine annealing, batch size 140, 2000 epochs

EPOCH	1:	LOSS train	7.6404e-01	valid	8.5070e-01	lr	2.0e-03	0.0s
EPOCH	201:	LOSS train	9.5637e-05	valid	6.5516e-05	lr	2.0e-03	6.2s
EPOCH	401:	LOSS train	4.0122e-03	valid	8.1863e-04	lr	1.8e-03	11.6s
EPOCH	601:	LOSS train	5.6911e-03	valid	1.1311e-03	lr	1.6e-03	17.2s
EPOCH	801:	LOSS train	1.1803e-04	valid	6.2159e-05	lr	1.3e-03	22.7s
EPOCH	1001:	LOSS train	8.4910e-05	valid	2.7833e-05	lr	1.0e-03	28.3s
EPOCH	1201:	LOSS train	5.8358e-05	valid	5.0959e-05	lr	6.9e-04	33.7s
EPOCH	1401:	LOSS train	4.8770e-05	valid	1.8962e-04	lr	4.1e-04	38.9s
EPOCH	1601:	LOSS train	1.2372e-05	valid	5.8839e-06	lr	1.9e-04	44.3s
EPOCH	1801:	LOSS train	7.7949e-06	valid	4.4264e-06	lr	4.9e-05	49.5s
EPOCH	2000:	LOSS train	6.6936e-06	valid	4.1915e-06	lr	1.0e-07	54.7s

finished in 54.7s on CPU
final MSE 6.6936e-06 (down 1.1e+05x from the first epoch)
best validation MSE 3.7361e-06 (weights saved to model_best.pt)

test mse 8.7829e-06

derivative L2 error 1.6757e+00
derivative relative L2 error 9.1841e-02